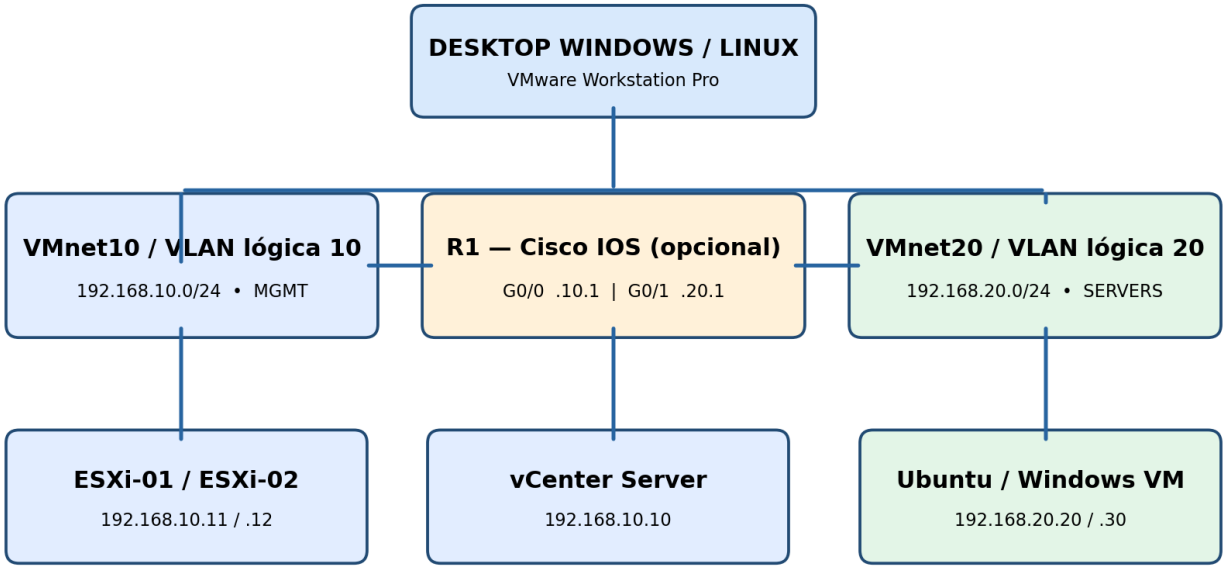
VMware vSphere + Cisco Networking

Certification Practice LAB | Installation, IP Plan, CLI, Validation & Troubleshooting

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Nota: VMnet10/20 são segmentos virtuais; não são VLANs 802.1Q automaticamente.

1. Scope and architecture

A desktop Windows/Linux runs VMware Workstation Pro. Nested ESXi-01 and ESXi-02 plus a vCenter Server Appliance are used for management exercises. Two isolated Workstation virtual networks (VMnet10 and VMnet20) represent management and workloads. A Cisco IOS router is an optional separate networking exercise: it must be run in an authorized simulator/emulator with interfaces connected to both segments. Workstation VMnet numbers are NOT automatically IEEE 802.1Q VLAN tags.

Important licensing: Workstation Pro is available at no cost; vCenter requires a valid entitlement or evaluation. Free standalone ESXi cannot be enrolled in vCenter. Check current Broadcom eligibility before downloading or deploying.

2. Host prerequisites and sizing

Component	Suggested allocation	Notes
Physical desktop	64 GB RAM minimum; 96 GB preferred	CPU VT-x/AMD-V, nested virtualization; NVMe 500 GB+
ESXi-01	2 vCPU, 8 GB RAM, 50 GB disk	Nested lab only; workloads require more

ESXi-02	2 vCPU, 8 GB RAM, 50 GB disk	Nested lab only
vCenter VCSA	Use vendor sizing wizard; budget ~14 GB RAM for Tiny	Exact requirements depend on vCenter release
Router VM	1 vCPU, 1–2 GB RAM	Cisco IOS image requires lawful license
Ubuntu VM	1–2 vCPU, 2 GB RAM	For network validation

3. Addressing plan

Device	Network / NIC	IPv4 / prefix	Gateway
R1 interface A	VMnet10	192.168.10.1/24	—
R1 interface B	VMnet20	192.168.20.1/24	—
vCenter	VMnet10	192.168.10.10/24	192.168.10.1
ESXi-01 vmk0	VMnet10	192.168.10.11/24	192.168.10.1
ESXi-02 vmk0	VMnet10	192.168.10.12/24	192.168.10.1
Ubuntu workload	VMnet20	192.168.20.20/24	192.168.20.1
Windows workload (optional)	VMnet20	192.168.20.30/24	192.168.20.1

Use a lab DNS resolver with forward/reverse records for vCenter and ESXi. Example FQDNs: vcsa.lab.example, esxi01.lab.example, esxi02.lab.example. Do not deploy vCenter without reliable DNS and time synchronization.

4. Create virtual networks in Workstation

1. Open Virtual Network Editor with administrative privileges. Add VMnet10 and VMnet20 as Host-only isolated networks. Disable DHCP if assigning all IPs statically.
2. VMnet10 subnet: 192.168.10.0/24; VMnet20 subnet: 192.168.20.0/24. Ensure no overlap with corporate/home VPN or LAN.
3. Attach ESXi management vNICs and vCenter to VMnet10. Attach the router interface A to VMnet10 and interface B to VMnet20.
4. For a basic workload, attach an Ubuntu VM directly to VMnet20. For nested workload inside ESXi, attach an additional ESXi vNIC to VMnet20 and configure a standard vSwitch + untagged port group on that uplink.
5. Enable nested hardware virtualization on ESXi VM settings; ensure your host OS and hypervisor expose VT-x/EPT or AMD-V/RVI.

5. Cisco IOS configuration — dual interfaces

This design uses two router interfaces, one per virtual segment, and therefore does not require 802.1Q subinterfaces. Interface names may differ in your IOS image.

```
enable
configure terminal
hostname R1-LAB
no ip domain-lookup
interface GigabitEthernet0/0
 description MGMT_VMnet10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface GigabitEthernet0/1
 description SERVER_VMnet20
 ip address 192.168.20.1 255.255.255.0
```

```
 no shutdown
exit
end
write memory
```

No static routes are required between directly connected networks. Do not add a default route unless you intentionally provide an upstream gateway.

6. ESXi installation and management

6. Create ESXi-01 and ESXi-02 virtual machines using compatible guest OS settings and installation ISO obtained through valid entitlement. Assign each VM a NIC on VMnet10 and an additional NIC on VMnet20 for nested workloads.
7. Install ESXi using the interactive installer; set a strong local administrator password. In DCUI (F2), configure Management Network with static addresses from the IP plan, gateway 192.168.10.1, DNS and hostnames.
8. From a workstation with VMnet10 access, open https://192.168.10.11/ui and https://192.168.10.12/ui. Accept certificates only after checking fingerprints in the lab.
9. On each ESXi, create a standard vSwitch using the uplink connected to VMnet20, and create a port group named PG-SERVERS with VLAN ID 0 (untagged). Attach nested workload VM NICs to PG-SERVERS.
10. Validate ESXi connectivity and time sync before deploying vCenter.

7. ESXi diagnostic CLI (SSH/ESXi Shell only when needed)

```
esxcli network ip interface ipv4 get
esxcli network ip route ipv4 list
esxcli network nic list
esxcli network vswitch standard list
esxcli network vswitch standard portgroup list
vmkping -I vmk0 192.168.10.1
vmkping -I vmk0 192.168.10.10
esxcli system time get
```

Prefer the ESXi Host Client for routine configuration. Enable SSH temporarily for troubleshooting and disable it after the exercise.

8. vCenter deployment and inventory

11. Run the VCSA installer from a supported Windows/Linux workstation and select Install.
12. Choose an ESXi deployment target and datastore, select appropriate appliance size and network VMnet10. Set static IP 192.168.10.10/24, gateway 192.168.10.1 and a functioning DNS resolver.
13. Complete appliance Stage 2, including SSO domain (e.g., vsphere.local), NTP and identity settings.
14. Access https://vcsa.lab.example/ui. Create Datacenter DC-LAB and Cluster CL-LAB, then add ESXi hosts if both are licensed/entitled for vCenter management.
15. Create or inspect standard vSwitches, port groups, host health and VM inventory. HA, DRS and vMotion require additional resources and suitable licensing; they are not enabled in this basic exercise.

9. Linux workload and test commands

```
# Ubuntu with NetworkManager (replace interface name as discovered)
```

```
nmcli device status
sudo nmcli con mod "Wired connection 1" ipv4.method manual ipv4.addresses 192.168.20.20/24 ipv4.gateway 192.168.20.1
sudo nmcli con up "Wired connection 1"
ip -br addr
ip route
ping -c 4 192.168.20.1
ping -c 4 192.168.10.11
traceroute 192.168.10.10
```

10. Cisco-style verification exercises

Test	Command	Expected
R1 interface state	show ip interface brief	G0/0 and G0/1 up/up
Connected networks	show ip route connected	192.168.10.0/24 and 192.168.20.0/24
ARP learning	show ip arp	Neighbor entries after traffic
R1 to ESXi	ping 192.168.10.11	Replies
R1 to Ubuntu	ping 192.168.20.20	Replies
ESXi to gateway	vmkping -I vmk0 192.168.10.1	Replies
Ubuntu to vCenter	ping -c 4 192.168.10.10	Replies if ICMP permitted
DNS	nslookup vcsa.lab.example	192.168.10.10
vSphere inventory	vSphere Client	Hosts connected and healthy

11. Troubleshooting scenarios

- Incorrect IP or prefix: compare host addresses against section 3; verify no overlapping VMnets.
- Router down/down: check emulator link mapping and virtual NIC attachment; run show interfaces and show ip interface brief.
- ESXi gateway unreachable: check vmk0 binding, physical adapter mapping to VMnet10 and vSwitch management port group.
- Workload cannot reach management: check VMnet20 uplink, router interface state, OS firewall and default route.
- vCenter deployment fails: verify forward/reverse DNS, NTP, adequate RAM, supported ESXi/vCenter compatibility and licensing.
- Nested VM has no network: confirm the additional ESXi NIC is connected to VMnet20 and the port group uses VLAN 0 for untagged traffic.

12. Practice assessment (certification checklist)

☐ Deploy both ESXi hosts and record management addresses.

☐ Create VMnet10 and VMnet20 and document isolation.

☐ Configure R1 routing and demonstrate inter-subnet reachability.

☐ Deploy vCenter with valid DNS/NTP and add licensed hosts.

☐ Create a workload port group and launch a test Linux VM.

☐ Capture evidence: screenshots, show commands, vmkping, DNS and inventory.

☐ Explain difference between Workstation VMnet, ESXi port group and IEEE 802.1Q VLAN.

13. References

Broadcom VMware documentation and licensing: https://knowledge.broadcom.com/ ; VMware vSphere documentation: https://techdocs.broadcom.com/ ; Cisco IOS command references: https://www.cisco.com/c/en/us/support/ios-nx-os-software/ios-software/products-command-reference-list.html

Educational note: Commands are illustrative and must be adapted to the specific ESXi, vCenter, Cisco IOS image and host OS versions. This is not an official Cisco or VMware certification lab.